

This is Exam 1 of MAT1120. You must show all work when required in order to receive credit and all work must be done independently. This exam is worth 100 points (roughly one third of your total grade thus far). You may consult anything in your text or posted to ASULearn. You may not consult anyone other than me. If I suspect academic dishonesty then the burden of proof is on you to demonstrate the material live. Academic dishonesty is included but not limited to: the use of techniques not covered yet, use of generative AI, and good old fashioned collaboration with peers. I will be available in Friday office hours 9:00 am EST to 9:50 am EST in zoom and EDH201A and again from 12:00 pm EST to 1:00 pm EST in EDH201A. If you need clarification on a problem feel free to contact me in office hours, else I will respond via email whenever I can. No late exams will be accepted without prior approval given solely at my discretion. Follow the directions of each problem. **Review your submission after turning in the exam and re-upload anything that might be difficult to parse/grade.**

1) Use the following information to answer parts a through d.

$$\int_{-2}^5 f(x) dx = 27 \quad \int_4^5 f(x) dx = 68 \quad \int_{-2}^5 g(x) dx = -15 \quad \int_4^5 g(x) dx = 33$$

a) Evaluate $\int_{-2}^5 (f(x) + g(x)) dx$. If you cannot evaluate it, tell me what information would be needed in order to do so.

$$\int_{-2}^5 f(x) dx = 27 \quad \int_{-2}^5 g(x) dx = -15$$

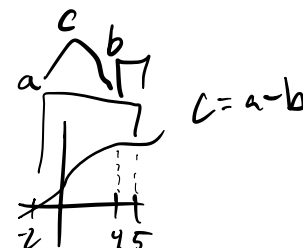
$$\int_{-2}^5 (f(x) + g(x)) dx = 27 + (-15) = 12$$

b) Evaluate $\int_{-2}^4 f(x) dx$. If you cannot evaluate it, tell me what information would be needed in order to do so.

$$\int_{-2}^4 f(x) dx = \int_{-2}^5 f(x) dx - \int_4^5 f(x) dx$$

$$27 - 68$$

$$27 - 68 = -41$$



c) Evaluate $\int_{-2}^4 (2g(x) - f(x)) dx$. If you cannot evaluate it, tell me what information would be needed in order to do so.

$$\int_{-2}^4 (2g(x) - f(x)) dx$$

$$\int_{-2}^4 g(x) dx = \int_{-2}^5 g(x) dx - \int_4^5 g(x) dx$$

$$-15 - 33 = -48$$

$$\int_{-2}^4 2g(x) dx = (-48) \cdot 2 = -96$$

$$\int_{-2}^4 (2g(x) - f(x)) dx = (-96) - (-41) = -96 + 41 = -55$$

d) Evaluate $\int_{-2}^6 f(x) dx$. If you cannot evaluate it, tell me what information would be needed in order to do so.

unable, would need information about $\int_5^6 f(x) dx$ as no information covers the range from 5 to 6 on $f(x)$

2a) Compute the following integral using a u -substitution. Refer to page 617 of the .pdf of our text for basic forms as needed.

$$\int \frac{5e^{\sqrt{x}}}{7\sqrt{x}} dx$$

$$u = \sqrt{x} = x^{\frac{1}{2}}$$

$$du = \frac{1}{2\sqrt{x}} dx \quad 2du = \frac{dx}{\sqrt{x}}$$

$$\frac{10}{7} e^{\sqrt{x}} + C$$

$$\int \frac{5e^{\sqrt{x}}}{7\sqrt{x}} dx = \frac{5}{7} \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$\frac{5}{7} \int e^{\sqrt{x}} \cdot \frac{dx}{\sqrt{x}}$$

$$\frac{5}{7} \int e^u \cdot 2du = \frac{10}{7} \int e^u du \nearrow$$

$$\frac{10}{7} e^u + C$$

2b) Compute the following integral using a u -substitution. Refer to page 617 of the .pdf of our text for basic forms as needed.

$$\int (18x^2 + 6e^x) \sin(x^3 + e^x + \pi) dx$$

$$u = x^3 + e^x + \pi$$

$$du = (3x^2 + e^x) dx$$

recall $\cos \theta + \pi = -\cos \theta$

$$6 \cos(x^3 + e^x) + C$$

$$(18x^2 + 6e^x) dx = 6(3x^2 + e^x) dx = 6 du$$

$$\int (6 du) \sin(u) dx$$

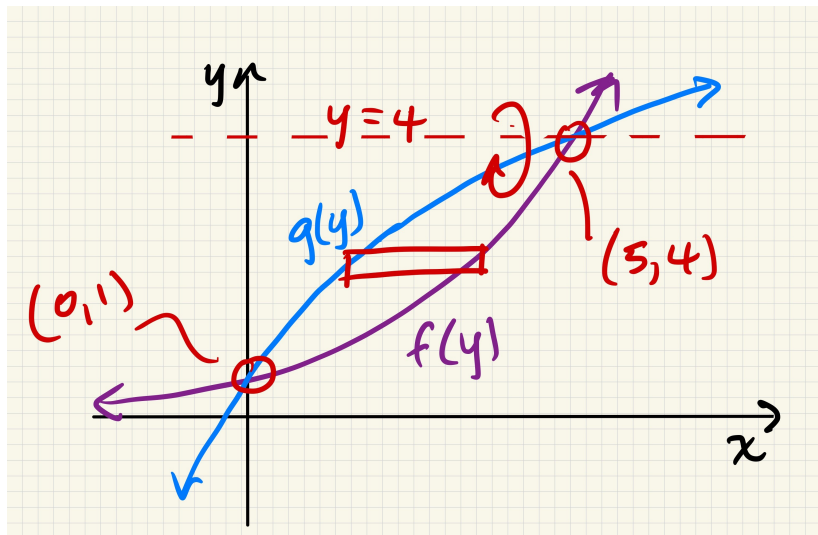
$$6 \int (du) \sin(u) \nearrow$$

Recall $\int \sin u du = -\cos u + C$ per p. 617

$$6 \int \sin u du = -6 \cos u + C$$

$$-6 \cos(x^3 + e^x + \pi) + C$$

3) Consider the follow diagram displaying a student's attempt at representing a solid of revolution. Based on the diagram answer the following questions.



a) Is this a disk/washer problem or a shell problem?

Shell, because the elements are parallel to the axis of rotation.

b) What is the axis of rotation?

The axis of rotation is the line $y=4$

c) Set up the integral to find the volume of the solid of revolution based on the diagram.

$$V = \int_c^d 2\pi R H \, dy$$

$$R = 4 - y$$

$$H = f(y) - g(y)$$

$$V = \int_1^4 2\pi (4-y)(f(y)-g(y)) \, dy$$

book convention would put 2π on the left side of $\int$

4a) Compute the following definite integral. Utilize symmetry as you feel appropriate.

$$\int_{-\pi}^{\pi} (\sin(x) + \cos(x) + e^x) dx$$

$$2 \int_0^{\pi} (\sin(x) + \cos(x) + e^x) dx$$

$$\int_0^{\pi} (\sin x + \cos x + e^x) dx = (1 + e^{\pi}) - (0)$$

$$2 \int_0^{\pi} (\sin x + \cos x + e^x) dx = 2(1 + e^{\pi}) - 2(0)$$

$$2 + 2e^{\pi}$$

$$\int (\sin(x) + \cos(x) + e^x) dx$$

$$-\cos x + \sin x + e^x$$

$$x = \pi$$

$$-\cos \pi + \sin \pi + e^{\pi}$$

$$-(-1) + 0 + e^{\pi}$$

$$1 + e^{\pi}$$

$$x = 0$$

$$-\cos 0 + \sin 0 + e^0$$

$$-(1) + 0 + 1 = 0$$

4b) Given that $f(x)$ is an even function and $g(x)$ is an odd function, rewrite the following integral in simplest terms. (Multiple answers are acceptable, do what you think is simplest to you.)

$$\int_{-a}^a e^4 f(x) - 8g(x) dx$$

even $f(-x) = f(x)$
odd $g(-x) = -g(x)$

$$e^4 \int_{-a}^a f(x) dx - 8 \int_{-a}^a g(x) dx$$

$$\int_0^a g(x) dx + \int_{-a}^0 g(x) dx = 0$$

$$g(x) + -g(x) = 0$$

$$8 \int_{-a}^a g(x) dx = 0$$

$$\int_0^a f(x) dx + \int_{-a}^0 f(x) dx = 2 \int_0^a f(x) dx$$

$$2e^4 \int_0^a f(x) dx$$

5) Find the area bounded by the functions $y = 8\sqrt{x}$ and $y = x^2$. You will need to find the bounds of integration. Report your answer as a fraction. (Note that the first function is eight times the square root of x and *not* the eighth root of x .)

Intersection

$$8\sqrt{x} = x^2$$

$$64x = x^4$$

$$x^4 - 64x = 0$$

$$x(x^3 - 64)$$

$x=0$ $x^3 - 64 = 0$
 $x=4$

$x=0$ & $x=4$

1) $y = 8\sqrt{x} = 8$ upper
 2) $y = x^2 = 16$ lower

$$\int_0^4 (8\sqrt{x} - x^2) dx$$

$$\int_0^4 (8x^{\frac{1}{2}} - x^2) dx$$

$$8\left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right) - \frac{x^3}{3}$$

$$\frac{16}{3}x^{\frac{3}{2}} - \frac{1}{3}x^3$$

$x=4$

$$\frac{16}{3}4^{\frac{3}{2}} - \frac{1}{3}4^3$$

$$\frac{16}{3}8 - \frac{64}{3}$$

$$\frac{128}{3} - \frac{64}{3}$$

$$\frac{64}{3}$$

$x=0$

$$\frac{16}{3}0^{\frac{3}{2}} - \frac{1}{3}0^3$$

$$0$$

$$\int_0^4 (8x^{\frac{1}{2}} - x^2) dx = \frac{64}{3}$$

6) Find the area bound by functions $y^2 = 4x$ and $4x - 3y - 4 = 0$. You may wish to first plot them and think about whether you want a dx or dy integral before proceeding.

$$x = \frac{y^2}{4} \text{ (left)} \quad x = \frac{3y+4}{4} \text{ (right)}$$

$$y^2 = 4x$$

$$y^2 - 3y - 4 = 0$$

$$(y-4)(y+1) = 0$$

$$y=4, y=-1$$

$$\int_{-1}^4 (x_R - x_L) dy = \int_{-1}^4 \left(\frac{3y+4}{4} - \frac{y^2}{4} \right) dy$$

$$\frac{1}{4} \int_{-1}^4 (-y^2 + 3y + 4) dy$$

$$\int (-y^2 + 3y + 4) dy = \left[-\frac{y^3}{3} + \frac{3y^2}{2} + 4y \right]_{-1}^4$$

$$y=4 \quad -\frac{4^3}{3} + \frac{3(4)^2}{2} + 4(4)$$

$$-\frac{64}{3} + \frac{3(16)}{2} + 16$$

$$-\frac{64}{3} + 40$$

$$\frac{56}{3}$$

$$y=-1 \quad -\frac{(-1)^3}{3} + \frac{3(-1)^2}{2} + 4(-1)$$

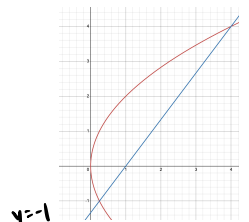
$$-\frac{1}{3} + \frac{3}{2} - 4$$

$$-\frac{13}{6}$$

$$\frac{56}{3} - \left(-\frac{13}{6} \right)$$

$$\frac{112}{6} + \frac{13}{6} = \frac{125}{6}$$

$$\frac{1}{4} \left(\frac{125}{6} \right) = \frac{125}{24}$$



7) Find the volume of a solid of revolution formed by rotating the region bounded by the graphs $y = x^2$ and $y = 0$ from $x = 0$ to $x = 2$ about the x -axis using the **disk** method.

$$V = \pi \int_a^b [f(x)]^2 dx$$

$$V = \pi \int_0^2 (x^2)^2 dx$$

$$V = \pi \int_0^2 x^4 dx$$

$$\int x^4 dx = \frac{x^5}{5}$$

$$V = \pi \left[\frac{x^5}{5} \right]_0^2 = \pi \left(\frac{2^5}{5} - \frac{0^5}{5} \right)$$

$$\pi \frac{32}{5}$$

$$\frac{32\pi}{5}$$

8) Find the volume of a solid of revolution formed by rotating region bounded by the graphs $y = x^2$ and $y = 0$ from $x = 0$ to $x = 2$ about the x -axis using the **shell** method. Your answers to #7 and #8 should agree. If not, try them again.

$$V = \int_0^4 2\pi y g(y) dy$$

$$V = 2\pi \int_0^4 R H dy$$

$$2\pi \int_0^4 y (2 - \sqrt{y}) dy$$

$$2\pi \int_0^4 y (2 - y^{\frac{1}{2}}) dy$$

$$2y - y^{\frac{3}{2}}$$

$$\int (2y - y^{\frac{3}{2}}) dy = \left(y^2 - \frac{2}{5} y^{\frac{5}{2}} \right)$$

$$\text{Right } x=2$$

$$\text{Left } x=\sqrt{y}$$

$$h = 2 - \sqrt{y}$$

$$y=4$$

$$(4)^2 - \frac{2}{5}(4)^{\frac{5}{2}}$$

$$16 - \frac{2}{5}(32)$$

$$16 - \frac{64}{5}$$

$$\frac{80}{5} - \frac{64}{5} = \frac{16}{5}$$

$$y=0 \quad 0^2 - \frac{2}{5} 0^{\frac{5}{2}}$$

$$0$$

$$V = 2\pi \left(\frac{16}{5} - 0 \right)$$

$$\frac{32\pi}{5}$$

9) Set up the integral to compute the arc length of graph $y = e^{-x^2}$ from $x = -1$ to $x = 1$. Do not attempt to solve your integral. Instead, use a computer program to evaluate your integral and report your answer to four decimal places of precision.

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = e^{-x^2} \cdot \frac{d}{dx}(-x^2) = -2xe^{-x^2}$$

$$\left(\frac{dy}{dx}\right)^2 = (-2xe^{-x^2})^2 = 4x^2e^{-2x^2}$$

$$\int_{-1}^1 \sqrt{1 + (4x^2e^{-2x^2})} dx$$

$$\approx 2.40888214175 \text{ or } \approx 2.4089$$

10) Set up the integral to compute the surface area of the shape created by rotating function $y = 5 \sin(x^3)$ from $x = 0$ to $x = 1$ about the x -axis. Do not attempt to solve your integral. Instead, use a computer program to evaluate your integral and report your answer to four decimal places of precision.

$$y = 5 \sin(x^3)$$

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = 5 \cos(x^3) \cdot \frac{d}{dx}(x^3) = 15x^2 \cos(x^3)$$

$$\left(\frac{dy}{dx}\right)^2 = (15x^2 \cos(x^3))^2 = 225x^4 \cos^2(x^3)$$

$$ds = \sqrt{1 + 225x^4 \cos^2 x^3} dx$$

$$\int_0^1 2\pi (5 \sin(x^3)) \sqrt{1 + 225x^4 \cos^2 x^3} dx$$

$$10\pi \int_0^1 \sin x^3 \sqrt{\quad} dx$$

$$S = \int_a^b 2\pi y ds$$

$$\approx 56.170089964$$

$$\approx 56.1700$$

Bonus Question: Create a slicing problem using isosceles right triangles as your cross sectional area in a region bounded by functions of your choosing so that the volume is 1 cubic-unit. You must write your problem out as if it were a test question to receive full credit as well as demonstrate your solution.